Operation Komorebi

Large Signal Model

A visual atlas of the finite foundations

Thirteen preserved mathematical illustrations.
State, transfer, probability and evidence.
No new empirical performance claims.

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Foundations 1.0.0: frozen finite theory release

Companion to PR-LASM software release 2.3.0

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Visual atlas of the finite foundations

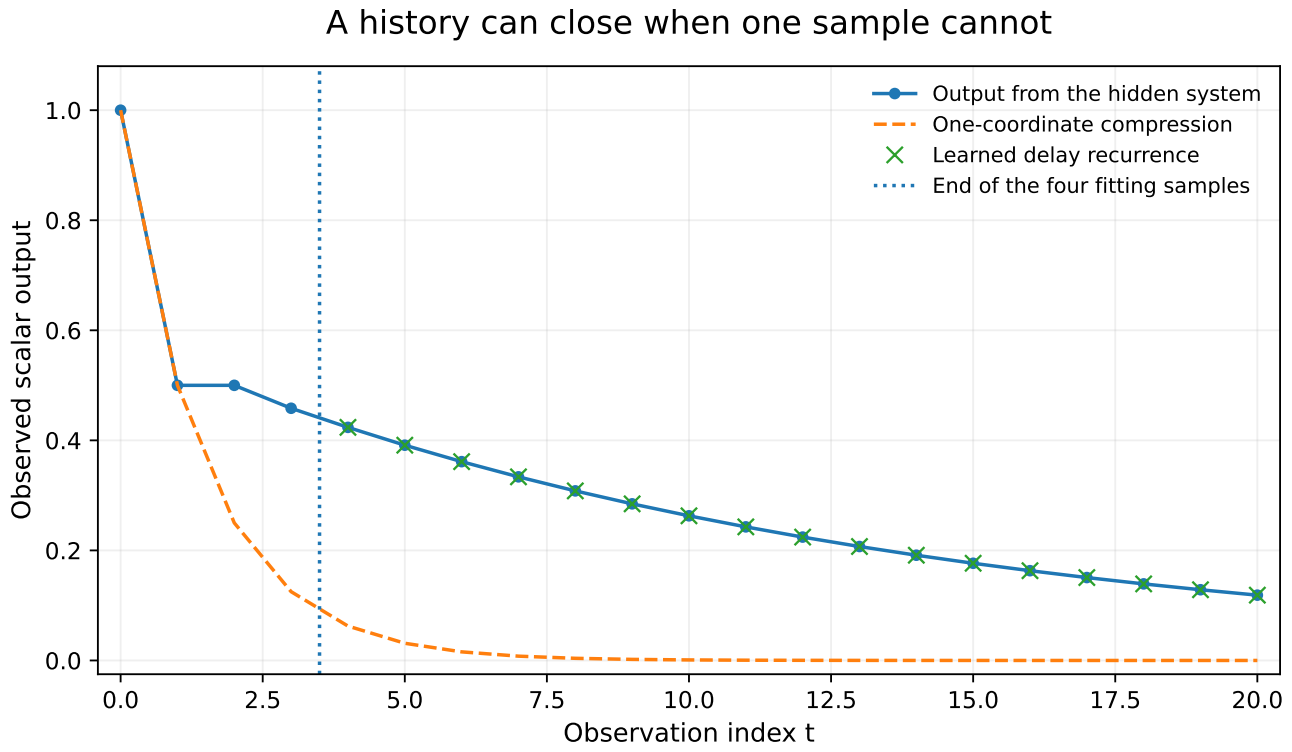
Foundations 1.0.0. Thirteen preserved mathematical illustrations.

This atlas follows the argument from observable state and packet interactions to admissible probability domains, continuous observations, repair and dependent-data confidence. Every plot is an unchanged Python-generated asset from the earlier foundations blocks. No new experiment, score or learned checkpoint is introduced.

Read the associated proposition and its setup before applying an illustrated result. Curves are finite calculations, certificate geometry or proved upper bounds as stated. A figure is not a proof of a quantified theorem. The source-identity ledger is `VISUAL_ATLAS_SOURCES.json`.

1. A history can carry the missing state

Theory: P14–17, P25.



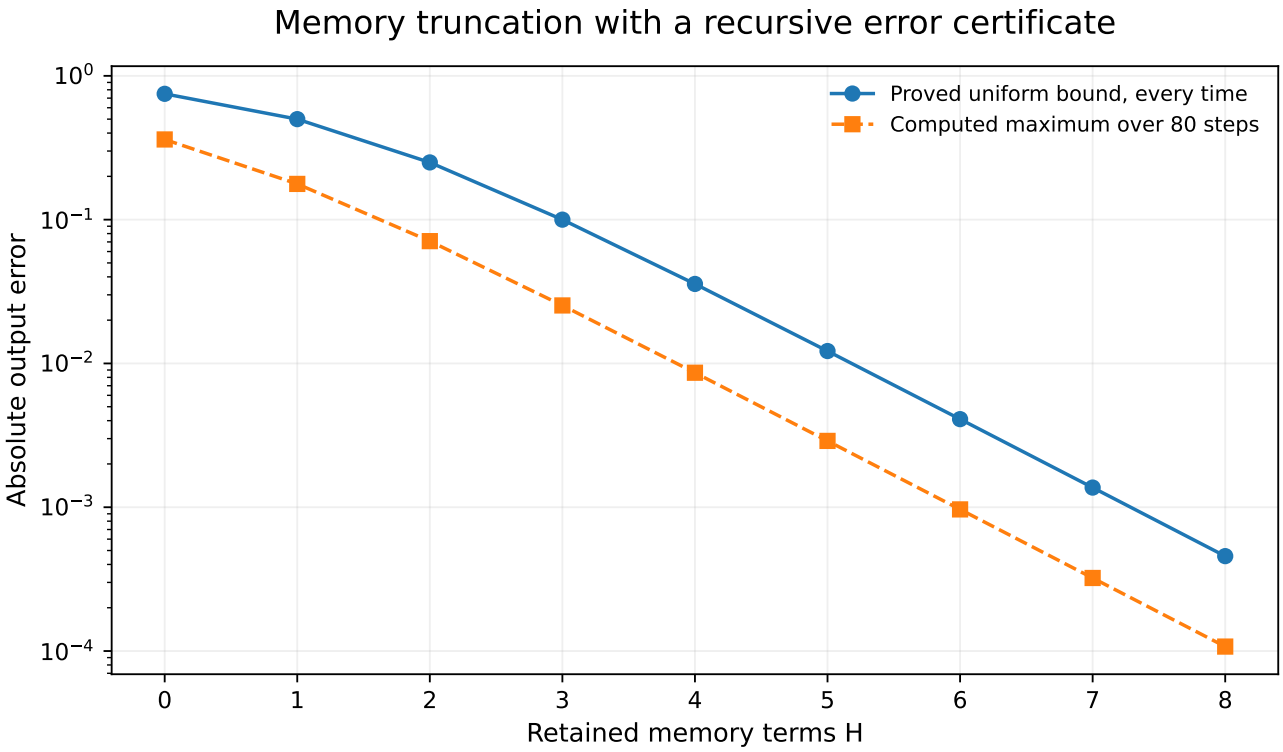
One scalar sample need not determine the next sample. A sufficiently long history can close under a declared finite linear model. The plotted delay realization uses the same output-only example as the cumulative volume. The vertical line separates the four fitting samples from continuation.

Boundary. The matching continuation is a finite example. Full identification depends on rank, delay closure and coverage, not on the appearance of these curves.

Preserved asset: `01_delay_prediction.pdf`.

2. Retain memory or enlarge the state

Theory: P10, P18–19.



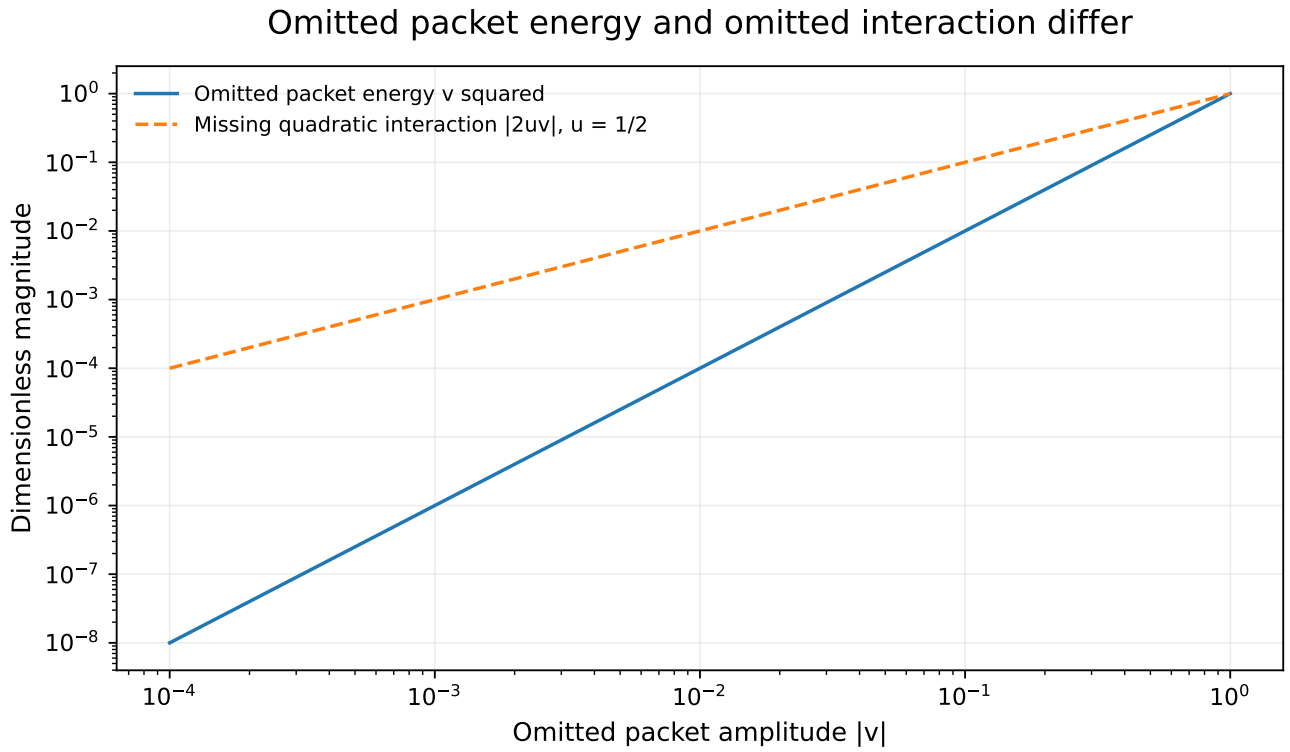
Eliminating unresolved coordinates creates an exact memory kernel. Truncating that kernel changes the dynamics. Under the stated contraction condition, the recursive prediction error has a computable upper bound. The plotted finite error and the bound are different quantities.

Boundary. The maximum computed over 80 steps does not establish the all-time statement. The written contraction argument supplies that statement under its hypotheses.

Preserved asset: `02_memory_certificate.pdf`.

3. Small omitted energy is not small influence

Theory: P5, P7, P20.



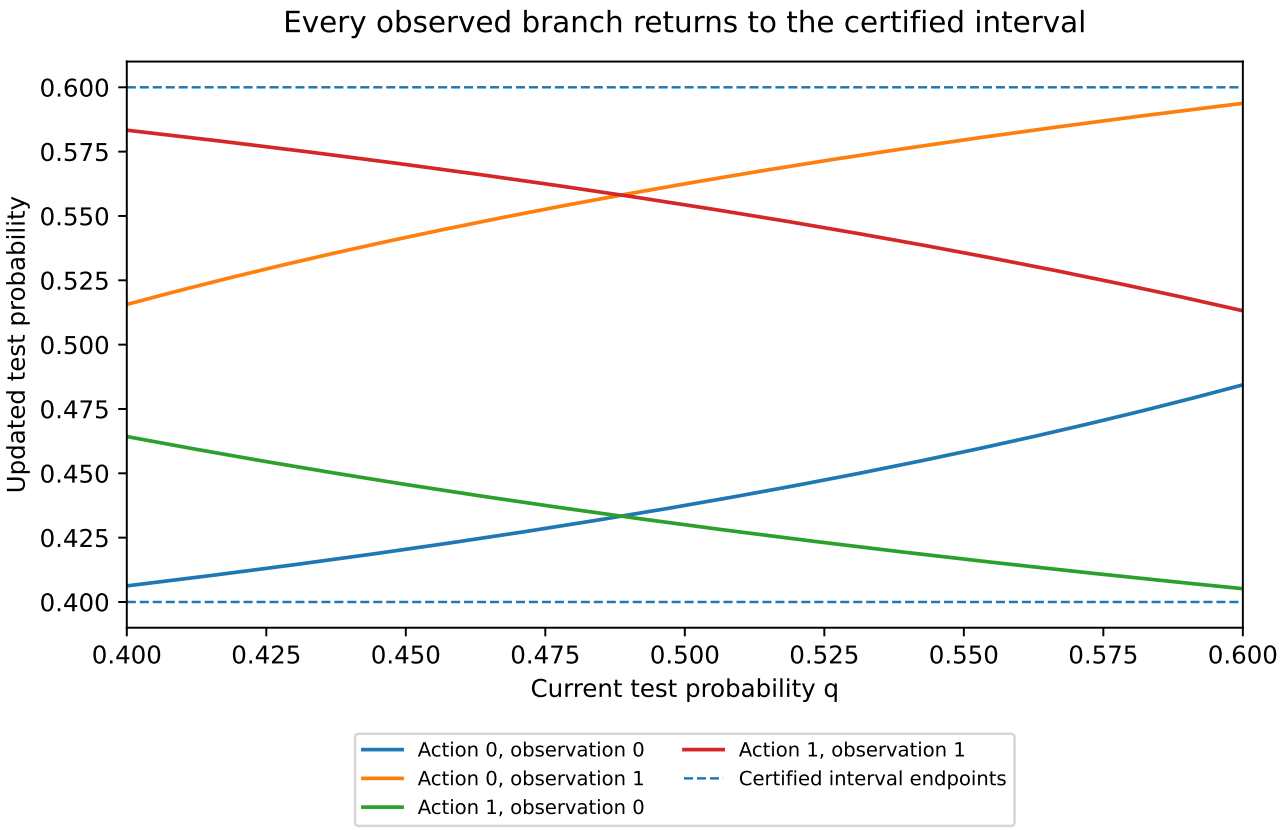
A nonlinear operation couples packets. For a square, the mixed term is proportional to the omitted amplitude while its energy is quadratic in that amplitude. The plot keeps those two scales visible. Refinement must account for transfer or interaction, not only reconstruction energy.

Boundary. A packet decomposition is not a collection of independent nonlinear subsystems. Dropping interaction terms requires its own approximation budget.

Preserved asset: `06_packet_interaction.pdf`.

4. A finite witness for all future branches

Theory: P39, P45, P49.



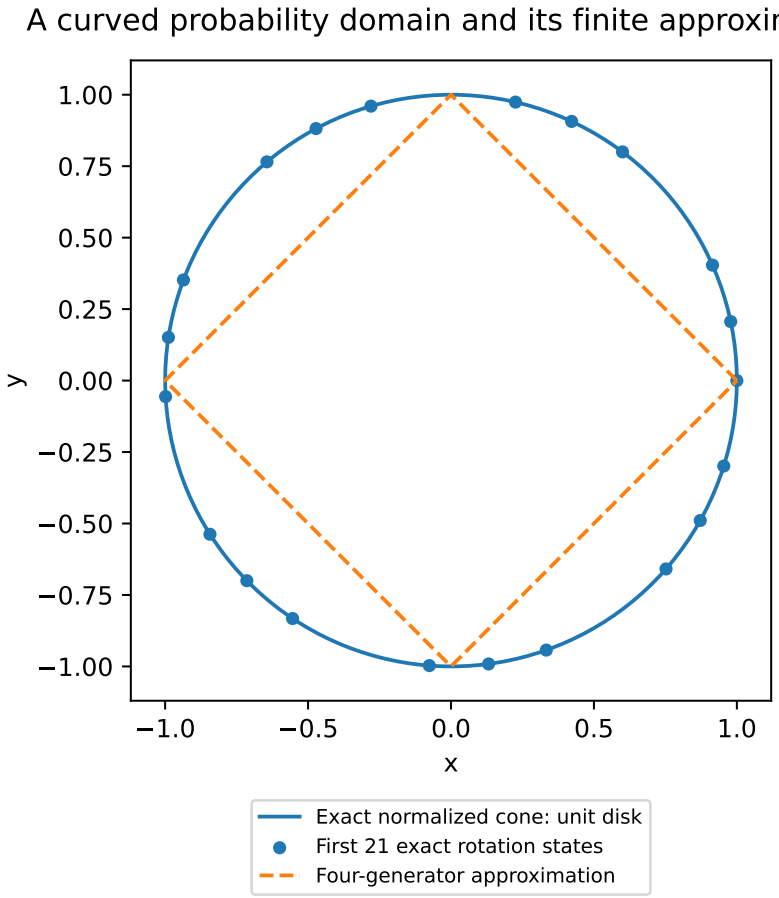
The fitted two-coordinate model maps every branch back into the interval from 2/5 to 3/5. Its exact positive-lift witness checks generators, nonnegative branch matrices, mass conservation and the initial weights. The plotted maps illustrate the same certified interval.

Boundary. The plotted grid is not the all-horizon proof. Intertwining and nonnegativity establish validity of the submitted model. They do not establish that it is the true process.

Preserved asset: `14_certified_interval.pdf`.

5. A small state may need a curved domain

Theory: P46–47.



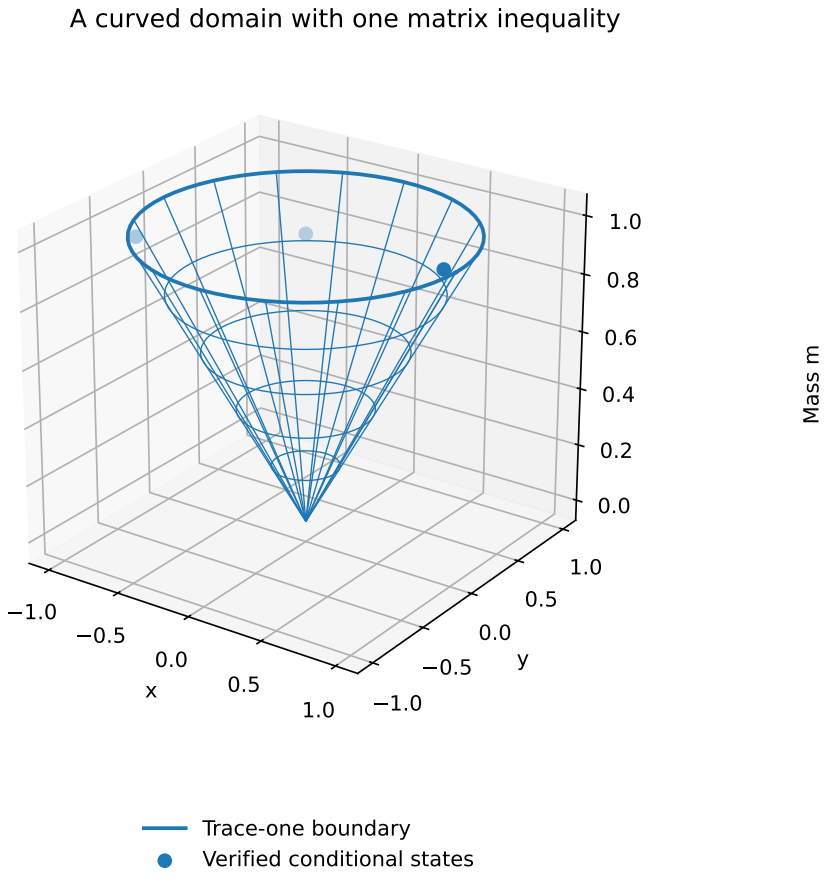
The rational rotation example has a three-coordinate realization whose normalized probability domain is the disk. The displayed orbit is finite, while the proof establishes density and rules out an exact finite polyhedral domain for this model.

Boundary. The inscribed diamond becomes useful only after a declared modification of the dynamics. Replacing the disk by a finite picture is not exact state refinement.

Preserved asset: `18_curved_probability_domain.pdf`.

6. A curved domain can have a finite certificate

Theory: P50–53.



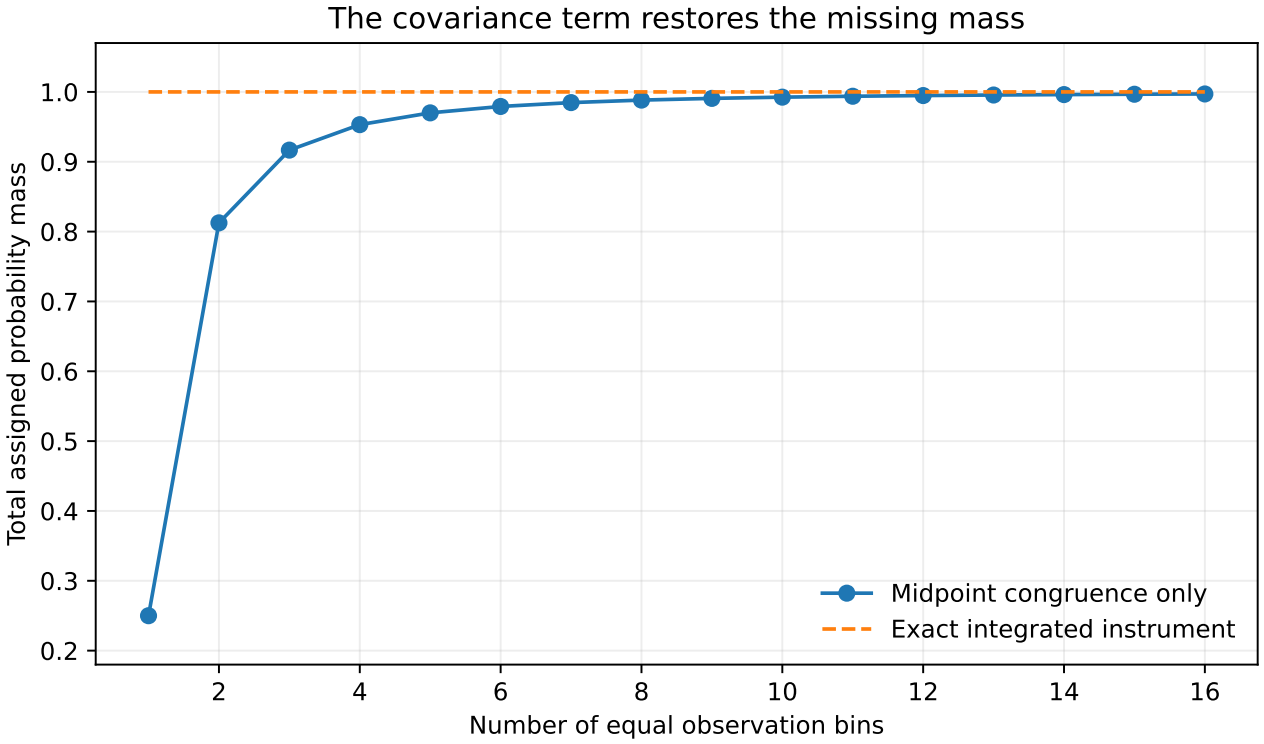
The Lorentz cone admits a two-by-two positive-semidefinite matrix representation. Congruence identities certify the unchanged rotation and observation operations. This resolves the certificate-language limitation without damping the exact model.

Boundary. The matrix representation is a mathematical positivity witness. No quantum hardware, physical quantum process or quantum advantage is asserted.

Preserved asset: `21_matrix_probability_cone.pdf`.

7. Averaging and quadratic action do not commute

Theory: P56–59.



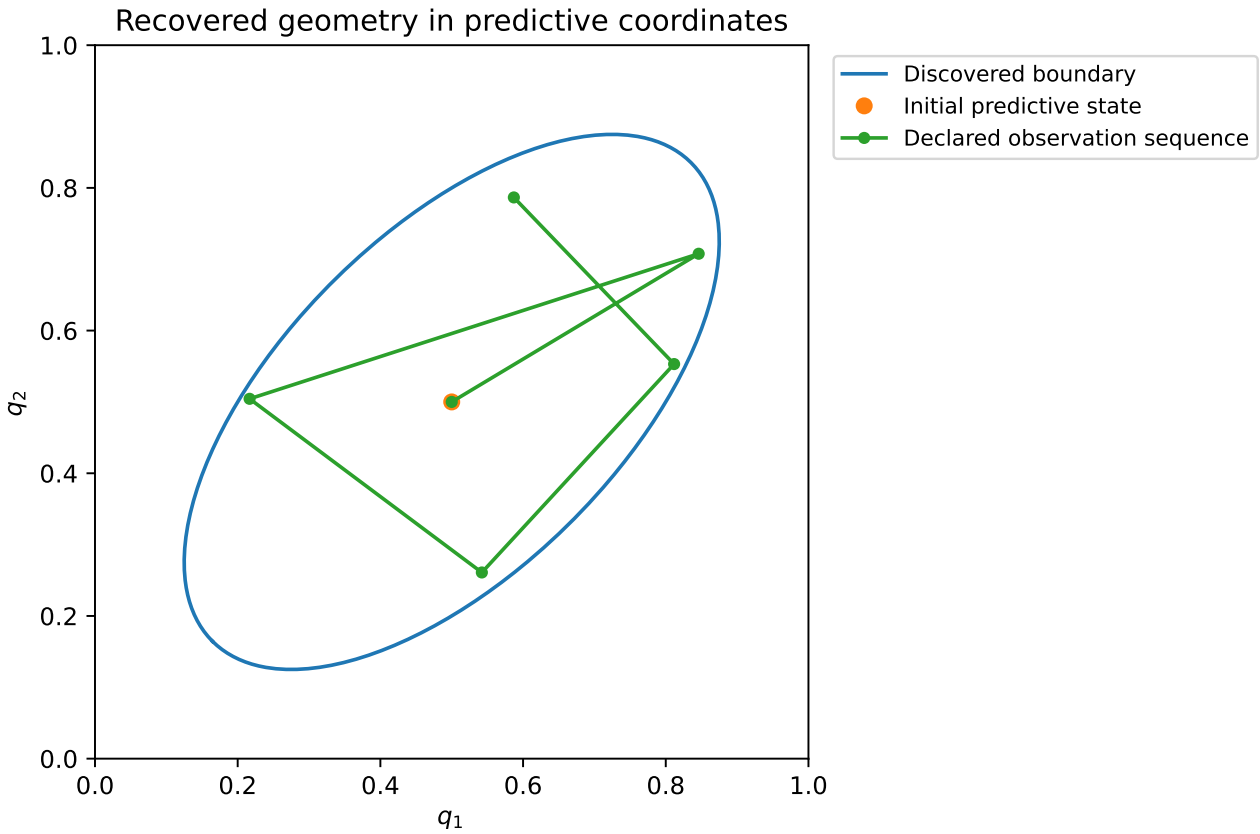
Midpoint congruences omit a covariance contribution. For this worked model, using n equal observation bins loses mass $3/(4n^2)$. Exact integration retains the missing term and preserves normalization. At one bin the naive approximation retains only one quarter of total mass.

Boundary. Event coarsening is exact for the stated bin-based policy. A policy that responds to values within a bin is a different comparison.

Preserved asset: `26_binning_mass_defect.pdf`.

8. Proposal and certificate are separate stages

Theory: P62–63.



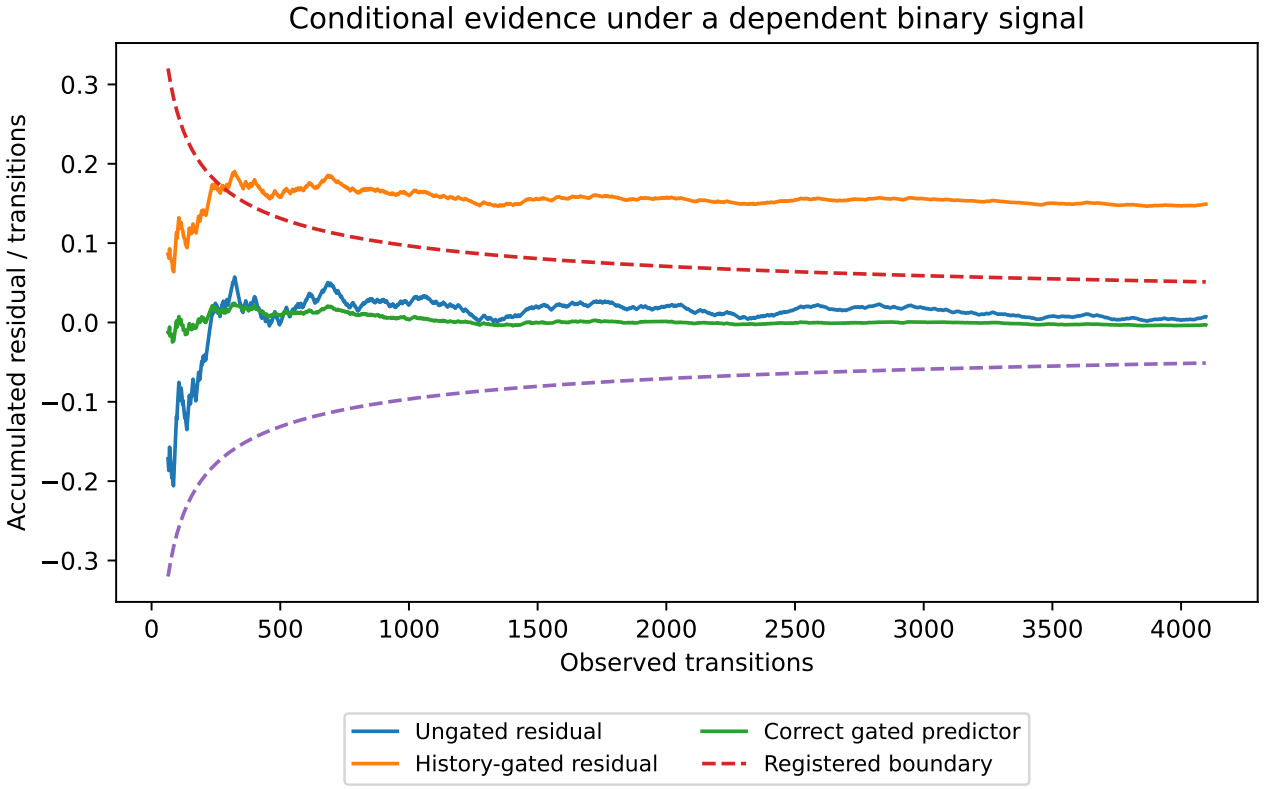
The exact example constructs an ellipse from observable operators, rather than taking its metric from a supplied template. The trajectory remains in the discovered domain. The accepted witness independently checks every operator, orientation and initial state.

Boundary. This is restricted exact invariant-form discovery, not general discovery of hidden state or noisy geometry from arbitrary observations.

Preserved asset: `28_discovered_geometry.pdf`.

9. Conditional tests can expose a missed dependence

Theory: P66–70.



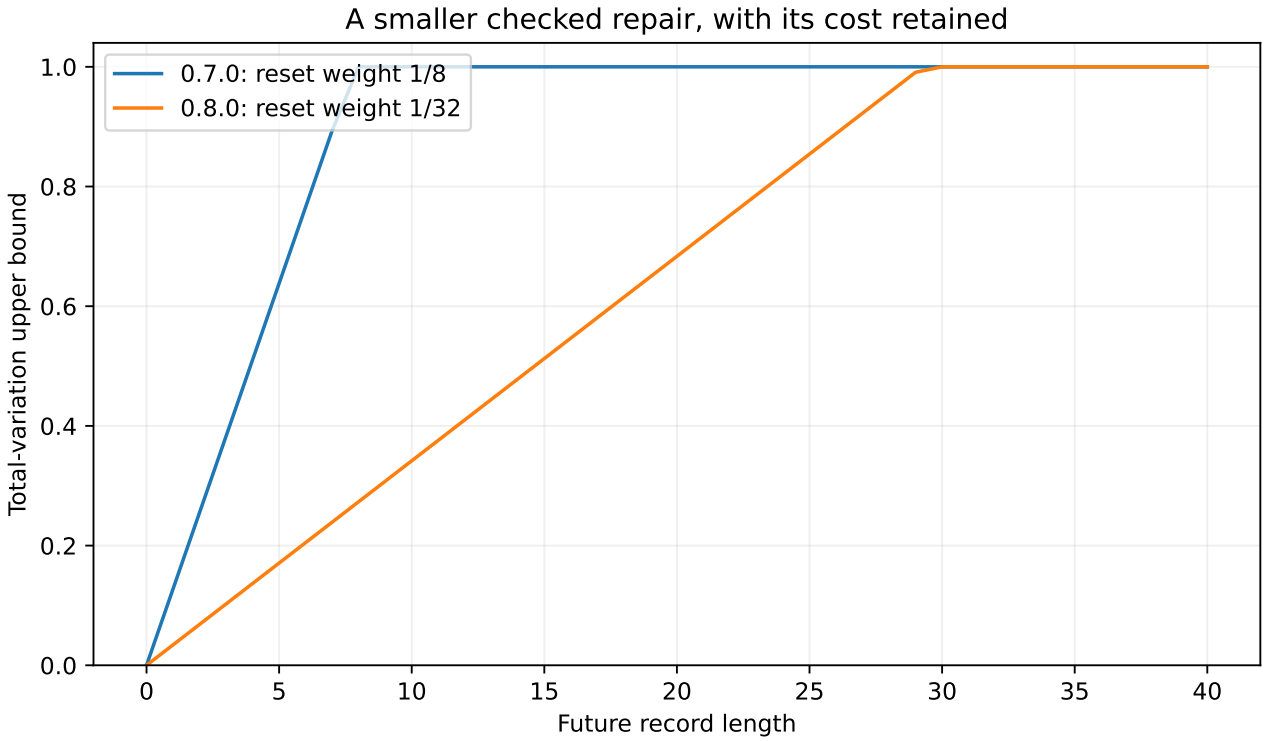
An ungated average can look compatible with a constant predictor while a prospectively chosen history-gated test reveals conditional mismatch. The figure uses the saved seeded path and its registered time-uniform boundary.

Boundary. It is one illustrative dependent path, not an empirical coverage study. Rejection can certify a lower discrepancy. Nonrejection is not an upper accuracy guarantee.

Preserved asset: `32_dependent_evidence.pdf`.

10. Validity repair has a measurable cost

Theory: P73, P79–80.



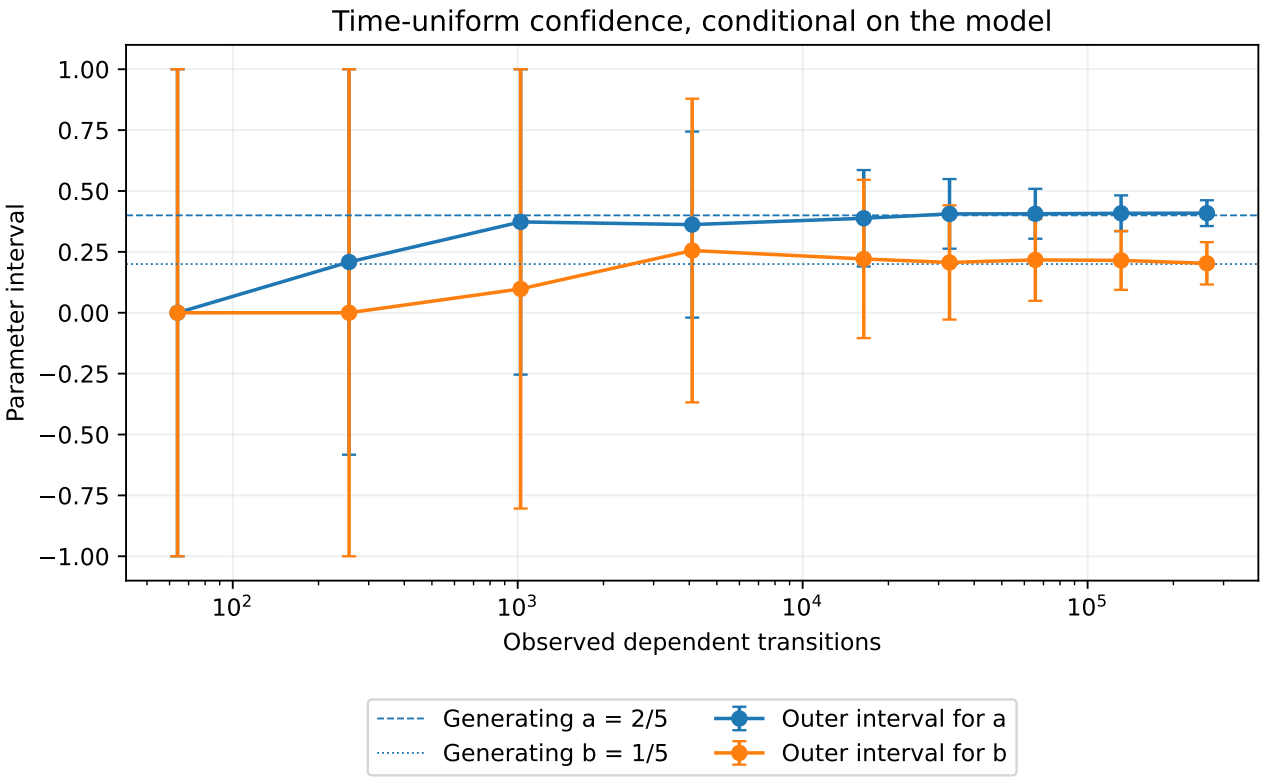
The same proposed geometry and perturbed operator family are certified with reset weight 1/32 instead of 1/8. The retained uncertainty radius is unchanged. The newer conservative full-record bound first becomes trivial at horizon 30 rather than eight.

Boundary. These lines are upper bounds, not observed model errors. The smaller repair is bracketed for one fixed geometry and reset path, not globally optimized.

Preserved asset: `41_smaller_repair.pdf`.

11. Learned coefficients need declared observations

Theory: P81–84, P87.



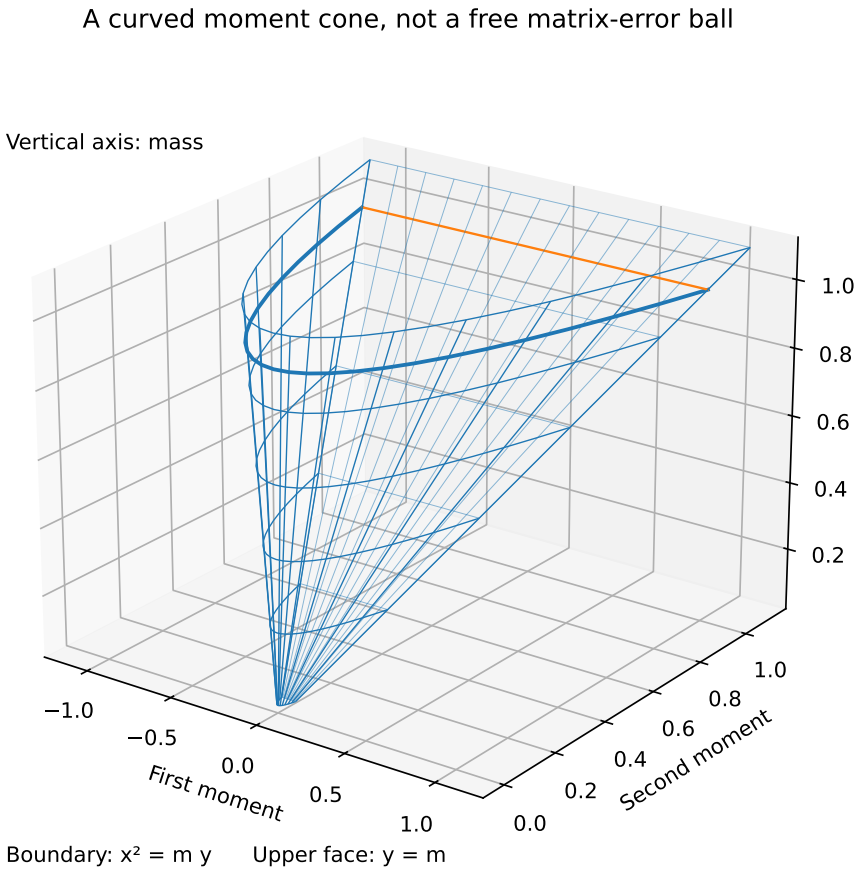
The intervals are from the saved five-node dependent path. The state map, two-parameter family, reference measure and predictable protocol are supplied. The learner estimates two parameters and rounds its intervals outward.

Boundary. The confidence guarantee is conditional and simultaneous over time. One saved path does not validate coverage empirically or authenticate the model assumptions.

Preserved asset: `42_confidence_intervals.pdf`.

12. Structured uncertainty can preserve a boundary ray

Theory: P83, P87.



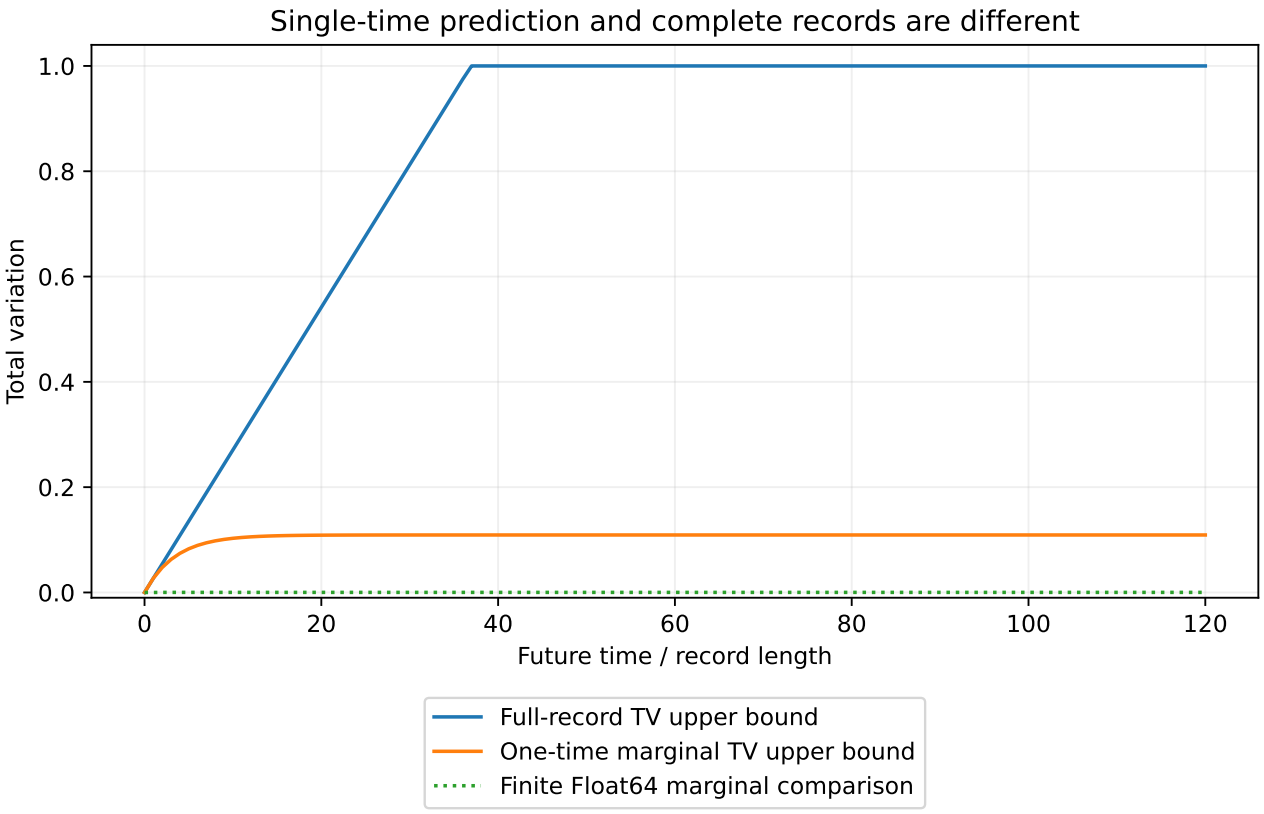
The moment cone is defined by a positive-semidefinite two-by-two moment matrix and an upper second-moment constraint. Parameter uncertainty changes the observation density but leaves the posterior state $v(u)=(1,u,u^2)$ unchanged.

Boundary. This structured parameter family is not a free matrix-entry uncertainty ball. The geometry and feature dictionary are declared, not discovered by this example.

Preserved asset: `44_moment_cone.pdf`.

13. Choose the guarantee that matches the question

Theory: P85–87.



The full-record bound grows with record length. The one-time marginal bound remains uniformly informative only because the declared Markov family has a common minorization. The figure also shows a finite numerical marginal comparison, separately from either theorem.

Boundary. An all-time single-marginal bound is not an all-time complete-record bound. Common reference measure, model class, initial state and policy remain part of the comparison.

Preserved asset: `45_prediction_horizons.pdf`.